



BAAZ JKBOSE

🔥 2026 CLASS 11th 🔥

SETS

MATHS

LECTURE-03

BY-MEHRAJ SIR



TOPICS TO BE COVERED

1 Types of Sets

2 Equal Sets

3 Exercise Discussion 1.2

Quick Recap

- * Roster form / Tabular form
- * Set-Builder form / Property Method.

$$A = \{2, 3, 5, 7, 11, 13\}$$

$$\Rightarrow A = \{x : x \text{ is a } \text{positive} \text{ prime no. } < 15\}$$

Types of a Sets

* Empty Set:

- A set which does not contain any element is known as an empty set.
- Also known as Null set or void set.
- Generally denoted by ϕ (phi) or $\{\}$

For example:

$$A = \{x : x \in \mathbb{N} \text{ \& } x < 1\} = \phi$$


$$B = \{x : x \in \mathbb{Z} \text{ \& } 1 < x < 2\} = \phi$$

$$C = \{x : x \in \mathbb{N} \text{ \& } x^2 - 2 = 0\} = \phi$$

$$x^2 - 2 = 0$$

$$\Rightarrow x^2 = 2$$

$$\Rightarrow x = \pm\sqrt{2} \notin \mathbb{N}$$

* Finite set:

A set which is either empty or contain definite number of elements is known as a finite set.

• $A = \{2, 4, 6, 8, 10\} \rightarrow$ finite set.

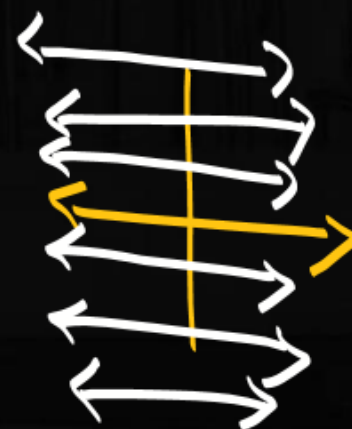
• $B = \{2, 4, 6, 8, \dots\} \rightarrow$ not a finite set (Infinite set)

* Infinite set:

A set which is not finite is known as an Infinite set.

eg; $\mathbb{N}, \mathbb{W}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}$
(Infinite sets)

* set of lines parallel to x-axis



~~finite set~~

Infinite Set ✓

Example

Which of the following sets are null sets:

- (a) Set of odd natural numbers divisible by 2. ϕ
- (b) Set of even prime numbers. $= \{2\} \neq \phi$ (Singleton set)
- (c) $\{x : x \text{ is a natural number, } x < 5 \text{ and } x > 7\} = \phi$
- (d) $\{y : y \text{ is a point common to any two parallel lines}\} = \phi$



Example

Which of the following sets are Finite or Infinite:

- (a) The set of months in a Year. (Finite set)
- (b) $\{1, 2, 3, \dots\}$ (Infinite set)
- (c) The set of positive integers greater than 100. (Infinite set)
- (d) The set of prime numbers less than 99. (Finite set)

Question

Cardinal number of an empty set is:

Cardinality

1

0 ✓

2

1 ✗

3

2 ✗

4

Infinite ✗

$\{1, 2, 3, 4, \dots\}$

Equal Sets

- Two sets A and B are said to be equal if they have exactly same elements

For example: $A = \{1, 2, 3\}$ and $B = \{3, 2, 1\}$
 $\Rightarrow A = B$

- $A = \{2, 3\}$ and $B = \{x : x \in \mathbb{N} \text{ \& } x^2 - 5x + 6 = 0\}$

Here $x^2 - 5x + 6 = 0 \Rightarrow x^2 - 2x - 3x + 6 = 0$

$$\Rightarrow x(x-2) - 3(x-2) = 0$$

$$\Rightarrow (x-2)(x-3) = 0$$

$$\Rightarrow x = 2, 3$$

$$\text{So, } B = \{x : x \in \mathbb{N} \text{ \& } x^2 - 5x + 6 = 0\}$$

$$\Rightarrow B = \{2, 3\}$$

$$\Rightarrow A = B$$

Q) If $n(A) = n(B)$, then $A = B$ *

(a) True / (b) false (✓)

Reason: $A = \{1, 2, 3\}$
and $B = \{4, 5, 6\}$

Here $n(A) = n(B)$
& $A \neq B$.

Q) If $A = B$, then $n(A) = n(B)$
(a) True / (b) false (✓)

* Equivalent sets :

Two sets A and B are said to be equivalent sets if $n(A) = n(B)$.

- * Equivalent sets may or may not be equal.
- * Equal sets are also equivalent sets.

Question

Value of (x, y) if the Set $A = \{2x - y, x - y\}$ and $B = \{3, 2\}$ are equal, is

- ☒ 1 $x = 1, y = -1$
- ☒ 2 $x = -1, y = -4$
- ☒ 3 Both 1 and 2 are incorrect
- ☒ 4 Both 1 and 2 are correct

Case 1st:

$$2x - y = 3 \quad \text{--- (1)}$$

$$x - y = 2 \quad \text{--- (2)}$$

Subtracting (1) & (2), we get

$$2x - y - x + y = 3 - 2$$

$$\Rightarrow x = 1$$

Also, $1 - y = 2$

$$\Rightarrow y = 1 - 2$$

$$\Rightarrow y = -1$$

$(x, y) = (1, -1)$

Case 2nd:

$$2x - y = 2 \quad \text{--- (3)}$$

$$x - y = 3 \quad \text{--- (4)}$$

Subtracting (3) & (4), we get:

$$2x - y - x + y = 2 - 3$$

$$\Rightarrow x = -1$$

So, $-1 - y = 3$

$$\Rightarrow y = -1 - 3 = -4$$

$(x, y) = (-1, -4)$

Question

Which of the following represents a pair of equal sets?

- 1 ☒ $A = \{1, 2, 3\}$ and $B = \{x : x \in \mathbb{N} \text{ and } x > 5\} = \{6, 7, 8, \dots\}$
- 2 ☒ $A = \{B, O, W, L\}$ and $B = \{x : x \text{ is a letter of the Word ELBOW}\} = \{E, L, B, O, W\}$
- 3 ☒ $A = \{1, 4, 9, 16\}$, $B = \{x : x = n^2, n \in \mathbb{N} \text{ and } 0 < n < 5\}$
 $= \{1, 4, 9, 16\}$
- 4 ☒ $A = \phi$ and $B = \{O, E, A\}$

Question

Which of the following is equal to set $A = \{x : x^2 + 2x + 24 > 0, x \in \mathbb{R}\}$

1 $B = \{6, 4\}$ Jan-Mehra 1729.

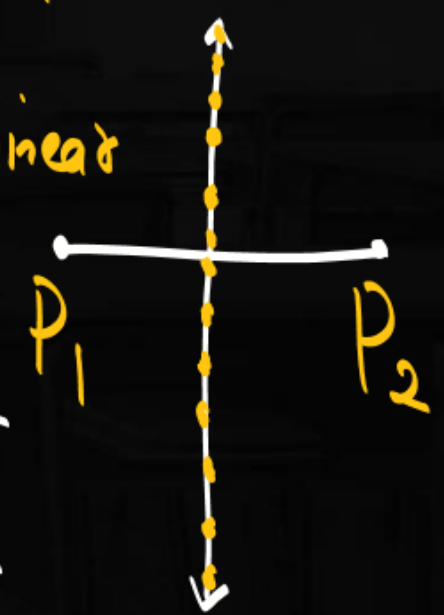
2 $\mathbb{R}$ (Homework.)

3 $B = \{-6, 4\}$

4 $\emptyset$

Q) no. of circles passing through 2 points in a plane is
~~(a) 1~~ ~~(b) 2~~ (c) Infinite. ~~(d) More than 2 but finite.~~

Q) no. of circles passing through 3 non-collinear points in a plane?
(a) 1 ~~(b) 2~~ ~~(c) Infinite.~~ ~~(d) More than 2 but finite.~~



EXERCISE – 1.2

Question

Which of the following are examples of the null set

- ☒ (i) Set of odd natural numbers divisible by 2 ϕ
- ☒ (ii) Set of even prime numbers $\neq (\phi)$
- ☒ (iii) $\{x : x \text{ is a natural numbers, } x < 5 \text{ and } x > 7\}$ ϕ
- ☒ (iv) $\{y : y \text{ is a point common to any two parallel lines}\}$ ϕ

Question

Which of the following sets are finite or infinite

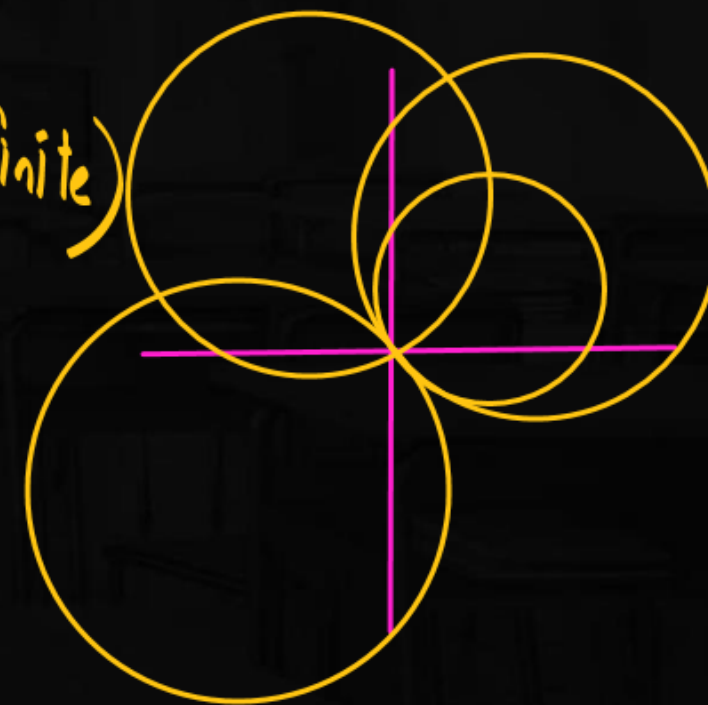
- ~~(i)~~ The set of months of a year ($\overline{f}$)
- ~~(ii)~~ $\{1, 2, 3, \dots\}$ ($\overline{f}$)
- ~~(iii)~~ $\{1, 2, 3, \dots, 99, 100\}$ (f)
- (iv) The set of positive integers greater than 100 ($\overline{f}$)
- ~~(v)~~ The set of prime numbers less than 99 (f)

Question

State whether each of the following set is finite or infinite:

- (i) The set of lines which are parallel to the x -axis (Infinite set)
- (ii) The set of letters in the English alphabet (Finite set)
- (iii) The set of numbers which are multiple of 5 $\longrightarrow \{5, 10, 15, 20, 25, \dots\}$ (Infinite)
- (iv) The set of animals living on the earth (Finite)
- (v) The set of circles passing through the origin $(0,0)$ (Infinite)

origin



Question

In the following, state whether $A = B$ or not:

- (i) $A = \{a, b, c, d\}$ $A = B$ $B = \{d, c, b, a\}$
- (ii) $A = \{4, 8, 12, 16\}$ $A \neq B$ $B = \{8, 4, 16, 18\}$ $\rightarrow \{2, 4, 6, 8, 10\}$
- (iii) $A = \{2, 4, 6, 8, 10\}$ $A = B$ $B = \{x : x \text{ is positive even integer and } x \leq 10\}$
- (iv) $A = \{x : x \text{ is a multiple of } 10\}$, $(A \neq B)$ $B = \{10, 15, 20, 25, 30, \dots\}$
 $\{10, 20, 30, \dots\}$

Question



Are the following pair of sets equal? Given reasons.

(i) $A = \{2, 3\},$

$B = \{x : x \text{ is solution of } x^2 + 5x + 6 = 0\}$

(ii) $A = \{x : x \text{ is a letter in the word FOLLOW}\}$

$B = \{y : y \text{ is a letter in the word WOLF}\}$

(Sol)(i) Here $x^2 + 5x + 6 = 0$

$$\Rightarrow x^2 + 2x + 3x + 6 = 0$$

$$\Rightarrow x(x+2) + 3(x+2) = 0$$

$$\Rightarrow (x+2)(x+3) = 0$$

$$\Rightarrow x = -2, -3.$$

So $A = \{2, 3\}$ & $B = \{-2, -3\}$

$$\Rightarrow A \neq B.$$

(ii) $A = \{F, O, L, W\}, B = \{W, O, L, F\}$

$$\Rightarrow A = B.$$

Question

From the sets given below, select equal sets:

$$A = \{2, 4, 8, 12\}, \quad B = \{1, 2, 3, 4\},$$

$$C = \{4, 8, 12, 14\}, \quad D = \{3, 1, 4, 2\}$$

$$E = \{-1, 1\}, \quad F = \{0, a\}$$

$$G = \{1, -1\}, \quad H = \{0, 1\}$$

4p/w.

Thank
You